HW1

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1 Writting Assignment

1.1

记投影中心 (小孔)O，建立圆心为 C, 圆上任意两点 P_1, P_2, 及其投影点 C', P_1', P_2'

根据透视原理，$\triangle OCP_1 \sim \triangle OC'P_1', \triangle OCP_2 \sim \triangle OC'P_2'$

$$\frac{OC}{OC'} = \frac{OP_1}{OP_1'} = \frac{OP_2}{OP_2'} \tag{1}$$

由于在圆中 $OP_1 = OP_2$, $OP_1' = OP_2'$, 故而圆投影结果仍为一个圆，disk 的投影形状是圆形。

1.2

Case 1: Horizontal Plane $y = 0$

Three different line directions in this plane are: $\mathbf{d}_1 = [1, 0, 0], \mathbf{d}_2 = [0, 0, 1], \mathbf{d}_3 = [1, 0, 1]$

For a direction vector $\mathbf{d} = [x, y, z]$, the vanishing point (x_{vp}, y_{vp}) is given by: $x_{vp} = \frac{f \times x}{z}, y_{vp} = \frac{f \times y}{z}$

For $\mathbf{d}_1$: vp_1 is at infinty

For $\mathbf{d}_2$: $x_{vp2} = \frac{f \times 0}{1} = 0$, $y_{vp2} = \frac{f \times 0}{1} = 0$, $vp_2 = (0, 0)$

For $\mathbf{d}_3$: $x_{vp3} = \frac{f \times 1}{1} = f$, $y_{vp3} = \frac{f \times 0}{1} = 0$, $vp_3 = (f, 0)$

Case 2: Vertical Plane $x = 0$

Three different line directions in this plane are: $\mathbf{d}_1 = [0, 1, 0], \mathbf{d}_2 = [0, 0, 1], \mathbf{d}_3 = [0, 1, 1]$

For $\mathbf{d}_1$: vp_1 is at infinty

For $\mathbf{d}_2$: $x_{vp2} = \frac{f \times 0}{1} = 0$, $y_{vp2} = \frac{f \times 0}{1} = 0$, $vp_2 = (0, 0)$

For $\mathbf{d}_3$: $x_{vp3} = \frac{f \times 0}{1} = 0$, $y_{vp3} = \frac{f \times 1}{1} = f$, $vp_3 = (0, f)$

1.3

Given a line with direction vector $\mathbf{d} = [x, y, z]$ in 3D space, the vanishing point of the line when projected onto the image plane can be found using the equations: $x_{vp} = \frac{f \times x}{z}$, $y_{vp} = \frac{f \times y}{z}$

where f is the effective focal length of the camera.

For a plane defined by the equation $Ax + By + Cz + D = 0$, z can be expressed in terms of x and y as:

$$z = -\frac{A}{C}x - \frac{B}{C}y - \frac{D}{C}$$

Given a direction vector $\mathbf{d}$ lying on this plane, its vanishing point can be determined by substituting the above expression for z into the equations for x_{vp} and y_{vp}:

$$x_{vp} = \frac{f \times x}{-\frac{A}{C}x - \frac{B}{C}y - \frac{D}{C}}, y_{vp} = \frac{f \times y}{-\frac{A}{C}x - \frac{B}{C}y - \frac{D}{C}}$$

These equations will give the vanishing points for all lines that have direction vectors $\mathbf{d}$ lying on the plane $Ax + By + Cz + D = 0$.

2 Programming Assignment

2.1 Object Attributes

2.1.1 Binarize

Given a gray image, threshold=128, output a binary image with 255 intensity.

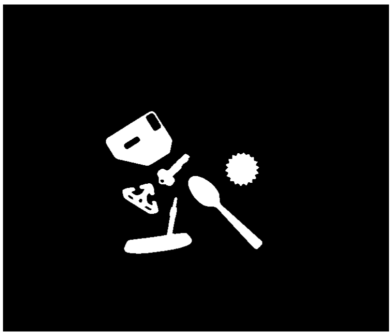

Figure 1: many objects 1 binary

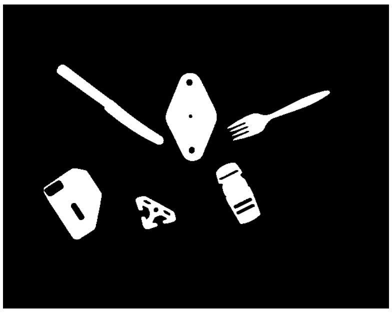

Figure 2: many objects 2 binary

2.1.2 label

Given a binary image, for every 1 pixel, traverse its 8 neighbors,.

If there is none labeled, assign a new label for it, else choose the min label of its neighbors, and remember that the other labels of its neighbors are equivalent to the min label.

When constructing the equivalent dictionary, make sure the min label isn't equivalent to any label, otherwise store the label equivalent to the min label.

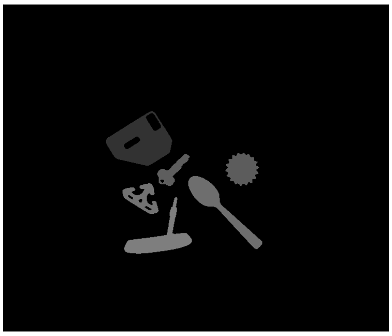

Figure 3: many objects 1 labeled

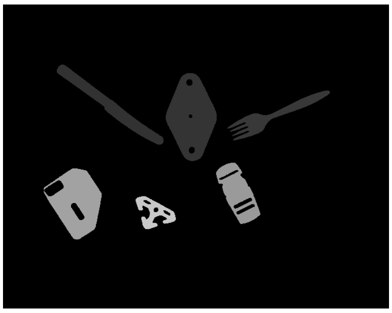

Figure 4: many objects 2 labeled

2.1.3 get attribute

Table 1: many objects 1

position	orientation	roundness
(266, 365)	0.08	0.52
(462, 313)	1.26	0.99
(326, 309)	0.77	0,13
(418, 241)	-0.77	0.02
(268, 258)	-0.54	0.49
(304, 178)	0.41	0.27

Table 2: many objects 2

position	orientation	roundness
(188, 358)	-0.64	0.01
(332, 338)	-1.53	0.31
(475, 340)	0.40	0.02
(414, 205)	-1.12	0.17
(130, 188)	-1.45	0.51
(266, 169)	-0.49	0.48

2.2 Hough Circles

2.3 Detect Edges

Sobel edge detector

$$Sobel_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -2 & 0 & 1 \end{bmatrix} \quad Sobel_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

See Figure 5

2.3.1 hough circles

edge threshold = 108

radius values=[20, 21, $\cdots$, 39, 40].

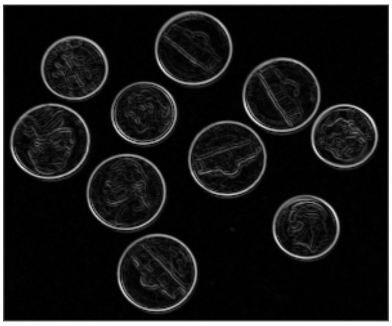

Figure 5: coins edges

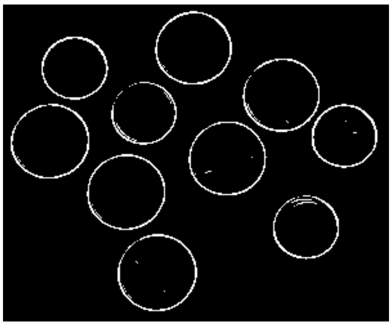

Figure 6: coins circle edges

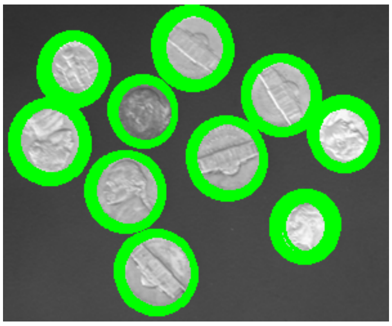

Figure 7: coins circle

I get accumulator array by counting per 10 degrees.
See Figure 6

2.3.2 find circles

hough threshold is assigned with the median of accumulator array.
See table 3 and Figure 7

r	x	y
31	31	147
25	47	54
29	68	215
26	81	108
26	99	264
30	104	35
30	117	173
29	144	94
26	170	234
30	206	118

Table 3: circles